Michael Soprano, PhD



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Education

Nov. 2019 – Jan. 2023 **PhD in Computer Science, Mathematics and Physics,** Cum Laude. 35th cycle, University of Udine, Italy.

Mar. 2015 – Mar. 2018 **Master's Degree in Computer Science,** 109/110, University of Udine, Italy.

Oct. 2011 – Mar. 2015 **Bachelor's Degree in Web Technologies and Multimedia,** 101/110, University of Udine, Italy.

Sep. 2006 – Jun. 2011 **High School Diploma** Head Technician Industrial Expert – Computer Science Specialization, 77/100, I.S.I.S. R. D'Aronco, Gemona del Friuli, Italy.

Work Experience

Jun. 2025 – Present **Postdoctoral Researcher (Research Contract Holder).** Department of Mathematics, Computer Science and Physics (DMIF), University of Udine, Italy.
Research contract pursuant to Article 22 of Law 240/2010, governed by the University regulation issued with Rector's Decree no. 308 of May 7, 2025. Research on information and disinformation, with focus on the automatic evaluation of information truthfulness, experimentation with detection models, and definition of evaluation metrics.

Mar. 2025 – May 2025 **Self-employed Worker.** Department of Economic and Statistical Sciences (DIES), University of Udine, Italy.
Design, implementation, and evaluation of a conversational system for accessing regional public healthcare services; prototype development, experimental evaluation, and data analysis within the inter-departmental project "Governance and Digital Public Administration."

Dec. 2023 – May 2025 **Postdoctoral Researcher.** DMIF, University of Udine.
PRIN 2022 project – "The Measure of Truth: An Evaluation-Centered Machine-Human Hybrid Framework for Assessing Information Truthfulness" (Code 20227F2ZN3, CUP G53D23002800006), funded by the EU – Next Generation EU – PNRR M4 C2 I1.1. Definition of an evaluation framework for misinformation-detection systems: creation of labelled datasets, bias prevention, and metrics that account for information truthfulness.

Mar. 2023 – Nov. 2023 **Postdoctoral Researcher.** DIES, University of Udine.
Research on eGovernment applications based on conversational agents for the Italian public administration, with particular emphasis on the healthcare domain.

Jan. 2022 – Mar. 2022 **Self-employed Worker.** Polytechnic Department of Engineering and Architecture (DPIA), University of Udine.
Development of a web application for the comparability and reproducibility of research results in combinatorial optimization.

Jun. 2018 – Oct. 2019 **Research Fellow.** DMIF, University of Udine.
Research project co-funded by SISSA Medialab: design, implementation, and evaluation of an ecosystem providing an alternative to traditional peer review in scholarly publishing.

Jun. 2017 – Oct. 2019 **Self-employed Worker.** Datamantix S.r.l., Udine, Italy.
Design, development, and maintenance of various web applications and a chatbot as a full-stack developer.

Aug. 2017 – Feb. 2018 **Self-employed Worker.** Archeido S.r.l., Udine, Italy.
Development of a web application for managing and visualizing financial assets as a full-stack developer.

Nov. 2015 – Feb. 2016 **Self-employed Worker.** Troublezine.
Design, development, and maintenance of a music news and events blog as a full-stack web developer.

Visits

Mar. 2022 – Jul. 2022 **Visiting Student**, Centrum Wiskunde & Informatica (CWI), Human-Centered Data Analytics Group, Amsterdam, The Netherlands. Research visit during the PhD programme.

Presentations

Oct. 2026 Public Outreach Workshop. **Intelligenza Artificiale e salute**. *Festival Collega-Menti 2026*, curated by Eddy Maddalena and Michael Soprano. October 4, 2026. Palazzo di Toppo Wassermann, Udine, Italy.
Programme: https://festivalcollegamenti.it/collegamenti2026/programma/45/intelligenza-artificiale-e-salute/
Guests: https://festivalcollegamenti.it/collegamenti2026/ospiti/

Sep. 2026 School Outreach Seminar. **A Caccia di Fake News**. *PCTO Summer Camp Informatica & Intelligenza Artificiale 2026*, University of Udine. September 10, 2026. Udine, Italy.
Slides: https://michaelsoprano.com/media/talks/PCTO-2026_A-Caccia-di-Fake-News.pdf

Jun. 2026 Workshop Talk. **Crowdsourced Fact-Checking for Misinformation: A Journey Through Nine Years of Research (2018–2026)**. *Workshop "AI and Disinformation", held as part of the 6th National CINI Conference on Artificial Intelligence (Ital-IA 2026)*, June 18, 2026. Rome, Italy.
Slides: https://michaelsoprano.com/media/talks/Ital-IA-2026_Crowdsourced-Fact-Checking-for-Misinformation-A-Journey-Through-Nine-Years-of-Research.pdf

Jun. 2026 Poster. **Crowdsourced Keypoint Matching for Region of Interest Identification in Cross-Stain Whole Slide Image Registration**. *22nd European Congress on Digital Pathology (ECDP 2026)*, June 18, 2026. Graz, Austria.
Poster PDF: https://michaelsoprano.com/media/posters/ECDP-2026_Crowdsourced-Keypoint-Matching-for-ROI-in-Cross-Stain-WSI-Registration.pdf

Apr. 2026 Workshop Talk. **Beyond Seeing Is Believing: On Crowdsourced Detection of Audiovisual Deepfakes**. *The 6th Workshop on Reducing Online Misinformation through Credible Information Retrieval (ROMCIR 2026), held as part of The 48th European Conference on Information Retrieval (ECIR 2026)*, April 2, 2026. Delft, The Netherlands.
Slides: https://michaelsoprano.com/media/talks/ROMCIR-2026_Beyond-Seeing-Is-Believing-On-Crowdsourced-Detection-of-Audiovisual-Deepfakes.pdf

Mar. 2026 Poster. **Analyzing AI Evaluation Benchmarks Through Information Retrieval and Network Science**. *The 48th European Conference on Information Retrieval (ECIR 2026)*, March 30, 2026. Delft, The Netherlands.
Poster PDF: https://michaelsoprano.com/media/posters/Analyzing-AI-Eval-Benchmarks-Through-IR_ECIR-2026.pdf

Mar. 2026 Poster. **Large Language Models as Assessors: On the Impact of Relevance Scales**. *The 48th European Conference on Information Retrieval (ECIR 2026)*, March 30, 2026. Delft, The Netherlands.
Poster PDF: https://michaelsoprano.com/media/posters/LLMs-As-Assessors_ECIR-2026.pdf

Sep. 2025 — School Outreach Seminar. **A Caccia di Fake News**. *PCTO Summer Camp Informatica & Intelligenza Artificiale 2025*, University of Udine. September 4, 2025. Udine, Italy.
Slides: https://michaelsoprano.com/media/talks/PCTO-2025_A-Caccia-di-Fake-News.pdf

Jul. 2025 — Poster. **AIDME: A Scalable, Interpretable Framework for AI-Aided Scoping Reviews**. *The 15th International Conference on Innovative Concepts and Theories in Information Retrieval*, July 18, 2025. Padua, Italy.
Poster PDF: https://michaelsoprano.com/media/posters/AIDME-Framework_ICTIR-2025.pdf

Jul. 2025 — Conference Talk. **The Magnitude of Truth: On Using Magnitude Estimation for Truthfulness Assessment**. *The 48th International ACM SIGIR Conference on Research and Development in Information Retrieval*, July 16, 2025. Padua, Italy.
Slides: https://michaelsoprano.com/media/talks/SIGIR-2025_The-Magnitude-of-Truth-On-Using-Magnitude-Estimation-for-Truthfulness-Assessment.pdf

Jun. 2025 — Workshop Talk. **AIDME: A Scalable, Interpretable Framework for AI-Aided Scoping Reviews**. *Workshop "AI, Misinformation and Disinformation", held as part of the 5th National CINI Conference on Artificial Intelligence (Ital-IA 2025)*, June 23, 2025. Trieste, Italy.
Slides: https://michaelsoprano.com/media/talks/Ital-IA-2025_AIDME-Framework.pdf

Feb. 2025 — Conference Talk. **Evaluation of Crowdsourced Peer Review using Synthetic Data and Simulations**. *21st Conference on Information and Research Science Connecting to Digital and Library Science (IRCDL 2025)*, February 20, 2025. Udine, Italy.
Slides: https://michaelsoprano.com/media/talks/IRCDL-2025_Evaluation-of-Crowdsourced-Peer-Review-using-Synthetic-Data-and-Simulations.pdf

Sep. 2024 — Demo Presentation. **Deploying Complex Crowdsourcing Tasks Without Fear: An Introduction to Crowd_Frame**. *The 3rd Italian Conference on Big Data and Data Science (ITADATA 2024)*, September 17, 2024. Pisa, Italy.
Slides: https://michaelsoprano.com/media/talks/ITADATA-2024_An-Introduction-To-Crowd_Frame.pdf

Oct. 2023 — Workshop Talk. **Towards a Conversational-Based Agent for Healthcare**. *CARE: Conversational Agents and Responsible Ethics in Healthcare Workshop, held as part of the European Federation of Medical Informatics Special Topic Conference 2023 (EFMI STC 2023)*, October 27, 2023. Turin, Italy.
Slides: https://michaelsoprano.com/media/talks/EFMI-STC-2023_Towards-a-Conversational-Based-Agent-for-Healthcare.pdf

May 2023 — Workshop Talk. **Towards a Conversational-Based Agent for Health Services**. *Workshop "AI for Public Administration", held as part of the 3rd National CINI Conference on Artificial Intelligence (Ital-IA 2023)*, May 29, 2023. Pisa, Italy.
Slides: https://michaelsoprano.com/media/talks/Ital-IA-2023_Towards-a-Conversational-Based-Agent-for-Health-Services.pdf

Feb. 2022 — Conference Talk. **Crowd_Frame: A Simple and Complete Framework to Deploy Complex Crowdsourcing Tasks Off-the-Shelf**. *The 15th ACM International Conference on Web Search and Data Mining (WSDM 2022)*, February 23, 2022. Phoenix, U.S.A. (remote).
Held remotely as a pre-recorded contribution due to the COVID-19 pandemic.
Slides: https://michaelsoprano.com/media/talks/WSDM-2022_Crowd_Frame-A-Simple-and-Complete-Framework-to-Deploy-Complex-Crowdsourcing-Tasks-Off-the-shelf.pdf
Video: https://dl.acm.org/doi/abs/10.1145/3488560.3502182

Oct. 2020 — Conference Talk. **The COVID-19 Infodemic: Can the Crowd Judge Recent Misinformation Objectively?**. *The 29th ACM International Conference on Information and Knowledge Management (CIKM 2020)*, October 19, 2020. Galway, Ireland (remote).
Held remotely as a pre-recorded contribution due to the COVID-19 pandemic.
Slides: https://michaelsoprano.com/media/talks/CIKM-2020_The-COVID-19-Infodemic.pdf
Video: https://dl.acm.org/doi/10.1145/3340531.3412048

Jul. 2020 — Conference Talk. **Can The Crowd Identify Misinformation Objectively? The Effects of Judgment Scale and Assessor's Background**. *The 43rd International ACM SIGIR Conference on Research and Development in Information Retrieval (SIGIR 2020)*, July 25, 2020. Xi'an, China (remote).
Held remotely as a pre-recorded contribution due to the COVID-19 pandemic.
Slides: https://michaelsoprano.com/media/talks/SIGIR-2020_Can-The-Crowd-Identify-Misinformation-Objectively.pdf
Video: https://dl.acm.org/doi/10.1145/3397271.3401112

Sep. 2019 — Workshop Talk. **Bias and Fairness in Effectiveness Evaluation by Means of Network Analysis and Mixture Model**. *Italian Information Retrieval Workshop (IIR 2019)*, September 16, 2019. Padua, Italy.
Slides: https://michaelsoprano.com/media/talks/IIR-2019_Bias-and-Fairness-in-Effectiveness-Evaluation-by-Means-of-Network-Analysis-and-Mixture-Model.pdf

Publications

Journal Articles

1. Francesca Da Ros, **Michael Soprano**, Luca Di Gaspero, and Kevin Roitero. «Large Language Models for Combinatorial Optimization: A Systematic Review». In: *ACM Computing Surveys* 58.11 (Aug. 2026). Journal Ranks: Journal Citation Reports (JCR) Q1 (2024), Scimago (SJR) Q1 (2024). ISSN: 0360-0300. DOI: 10.1145/3801961. Preprint: https://arxiv.org/abs/2507.03637.

2. Eddy Maddalena, Stefano Mizzaro, Kevin Roitero, Marco Viviani, David La Barbera, Sandip Modha, Gabriella Pasi, Francesca Da Ros, and **Michael Soprano**. «Report on the 14th Italian Information Retrieval Workshop (IIR 2024)». In: *SIGIR Forum* 58.2 (Dec. 2024), pp. 1–13. ISSN: 0163-5840. DOI: 10.1145/3722449.3722464.

3. Damiano Spina, Kevin Roitero, Stefano Mizzaro, Vincenzo Della Mea, Francesca Da Ros, **Michael Soprano**, Hafsa Akebli, Alex Falcon, Mehdi Fasihi, Alessio Fiorin, David La Barbera, Daniele Lizzio Bosco, Riccardo Lunardi, Alberto Marturano, Zaka-Ud-Din Muhammad, Francesco Nascimben, Moritz Nottebaum, Massimiliano Pascoli, Mihai Horia Popescu, Laura Rasotto, Mubashara Rehman, Francesco Taverna, Biagio Tomasetig, and Alessandro Tremamunno. «Report on the Hands-On PhD Course on Responsible AI from the Lens of an Information Access Researcher». In: *SIGIR Forum* 58.2 (Dec. 2024), pp. 1–61. ISSN: 0163-5840. DOI: 10.1145/3722449.3722456.

4. **Michael Soprano**, Kevin Roitero, Ujwal Gadiraju, Eddy Maddalena, and Gianluca Demartini. «Longitudinal Loyalty: Understanding The Barriers To Running Longitudinal Studies On Crowdsourcing Platforms». In: *ACM Transactions on Social Computing* 7.1–4 (Sept. 2024). Journal Rank: Scimago (SJR) Q2 (2023). DOI: 10.1145/3674884.

5. David La Barbera, Eddy Maddalena, **Michael Soprano**, Kevin Roitero, Gianluca Demartini, Davide Ceolin, Damiano Spina, and Stefano Mizzaro. «Crowdsourced Fact-checking: Does It Actually Work?» In: *Information Processing & Management* 61.5 (July 2024). Journal Ranks: Journal Citation Reports (JCR) Q1 (2023), Scimago (SJR) Q1 (2023), p. 103792. ISSN: 0306-4573. DOI: 10.1016/j.ipm.2024.103792.

6. **Michael Soprano**, Kevin Roitero, David La Barbera, Davide Ceolin, Damiano Spina, Gianluca Demartini, and Stefano Mizzaro. «Cognitive Biases in Fact-Checking and Their Countermeasures: A Review». In: *Information Processing & Management* 61.3 (2024). Journal Ranks: Journal Citation Reports (JCR) Q1 (2022), Scimago (SJR) Q1 (2022), p. 103672. ISSN: 0306-4573. DOI: 10.1016/j.ipm.2024.103672.

7. Kevin Roitero, David La Barbera, **Michael Soprano**, Gianluca Demartini, Stefano Mizzaro, and Tetsuya Sakai. «How Many Crowd Workers Do I Need? On Statistical Power When Crowdsourcing Relevance Judgments». In: *ACM Transactions on Information Systems* 42.1 (May 2023). Journal Ranks: Journal Citation Reports (JCR) Q1 (2021), Scimago (SJR) Q1 (2021). ISSN: 1046-8188. DOI: 10.1145/3597201.

8. Erik Brand, Kevin Roitero, **Michael Soprano**, Afshin Rahimi, and Gianluca Demartini. «A Neural Model to Jointly Predict and Explain Truthfulness of Statements». In: *Journal of Data and Information Quality*

15.1 (Dec. 2022). Journal Ranks: Journal Citation Reports (JCR) Q3 (2021), Scimago (SJR) Q2 (2021). ISSN: 1936-1955. DOI: 10.1145/3546917.

9. Davide Ceolin, Giuseppe Primiero, **Michael Soprano**, and Jan Wielemaker. «Transparent Assessment of Information Quality of Online Reviews Using Formal Argumentation Theory». In: *Information Systems* 110 (July 2022). Journal Ranks: Journal Citation Reports (JCR) Q2 (2021), Scimago (SJR) Q1 (2021), p. 102107. ISSN: 0306-4379. DOI: 10.1016/j.is.2022.102107.

10. **Michael Soprano**, Kevin Roitero, David La Barbera, Davide Ceolin, Damiano Spina, Stefano Mizzaro, and Gianluca Demartini. «The Many Dimensions of Truthfulness: Crowdsourcing Misinformation Assessments on a Multidimensional Scale». In: *Information Processing & Management* 58.6 (Nov. 2021). Journal Ranks: Journal Citation Reports (JCR) Q1 (2021), Scimago (SJR) Q1 (2021), p. 102710. ISSN: 0306-4573. DOI: 10.1016/j.ipm.2021.102710. Preprint: https://arxiv.org/abs/2108.01222.

11. Kevin Roitero, **Michael Soprano**, Beatrice Portelli, Massimiliano De Luise, Damiano Spina, Vincenzo Della Mea, Giuseppe Serra, Stefano Mizzaro, and Gianluca Demartini. «Can the Crowd Judge Truthfulness? A Longitudinal Study on Recent Misinformation About COVID-19». In: *Personal and Ubiquitous Computing* 27.1 (Sept. 2021). Journal Ranks: Journal Citation Reports (JCR) Q2 (2020), Scimago (SJR) Q1 (2021), pp. 59–89. ISSN: 1617-4917. DOI: 10.1007/s00779-021-01604-6. Preprint: https://arxiv.org/abs/2107.11755.

12. Kevin Roitero, **Michael Soprano**, Andrea Brunello, and Stefano Mizzaro. «Reproduce and Improve: An Evolutionary Approach to Select a Few Good Topics for Information Retrieval Evaluation». In: *Journal of Data and Information Quality* 10.3 (Sept. 2018). Journal Rank: Scimago (SJR) Q2 (2018). ISSN: 1936-1955. DOI: 10.1145/3239573.

Conference and Workshop Papers

13. Gaia Simeoni, **Michael Soprano**, Riccardo Lunardi, Kevin Roitero, and Stefano Mizzaro. «Analyzing AI Evaluation Benchmarks Through Information Retrieval and Network Science». In: *Advances in Information Retrieval: 48th European Conference on Information Retrieval (ECIR 2026)*. Vol. 16484. Lecture Notes in Computer Science. Conference Rank: CORE A. Cham: Springer Nature Switzerland, Mar. 2026, pp. 349–359. ISBN: 978-3-032-21300-6. DOI: 10.1007/978-3-032-21300-6_25.

14. Riccardo Zamolo, Riccardo Lunardi, **Michael Soprano**, Gianluca Demartini, Stefano Mizzaro, and Kevin Roitero. «Large Language Models as Assessors: On the Impact of Relevance Scales». In: *Advances in Information Retrieval: 48th European Conference on Information Retrieval (ECIR 2026)*. Vol. 16484. Lecture Notes in Computer Science. Conference Rank: CORE A. Cham: Springer Nature Switzerland, Mar. 2026, pp. 338–348. ISBN: 978-3-032-21300-6. DOI: 10.1007/978-3-032-21300-6_24.

15. **Michael Soprano**, Sandip Modha, Kevin Roitero, Eddy Maddalena, Marco Viviani, Gabriella Pasi, and Stefano Mizzaro. «AIDME: A Scalable, Interpretable Framework for AI-Aided Scoping Reviews». In: *Proceedings of the 2025 International ACM SIGIR Conference on Innovative Concepts and Theories in Information Retrieval (ICTIR)*. ICTIR '25. Padua, Italy: Association for Computing Machinery, July 2025, pp. 194–207. ISBN: 9798400718618. DOI: 10.1145/3731120.3744586.

16. Riccardo Lunardi, **Michael Soprano**, Paolo Coppola, Vincenzo Della Mea, Stefano Mizzaro, and Kevin Roitero. «PILs of Knowledge: A Synthetic Benchmark for Evaluating Question Answering Systems in Healthcare». In: *Proceedings of the 48th International ACM SIGIR Conference on Research and Development in Information Retrieval*. SIGIR '25. Conference Ranks: GGS A++, CORE A*. Padua, Italy: Association for Computing Machinery, July 2025, pp. 3648–3658. ISBN: 9798400715921. DOI: 10.1145/3726302.3730283.

17. Kevin Roitero, Dustin Wright, **Michael Soprano**, Isabelle Augenstein, and Stefano Mizzaro. «Efficiency and Effectiveness of LLM-Based Summarization of Evidence in Crowdsourced Fact-Checking». In: *Proceedings of the 48th International ACM SIGIR Conference on Research and Development in Information Retrieval*. SIGIR '25. Conference Ranks: GGS A++, CORE A*. Padua, Italy: Association for Computing Machinery, July 2025, pp. 457–467. ISBN: 9798400715921. DOI: 10.1145/3726302.3729960. Preprint: https://arxiv.org/abs/2501.18265.

18. **Michael Soprano**, Denis Eduard Tapu, David La Barbera, Kevin Roitero, and Stefano Mizzaro. «The Magnitude of Truth: On Using Magnitude Estimation for Truthfulness Assessment». In: *Proceedings of the*

48th International ACM SIGIR Conference on Research and Development in Information Retrieval. SIGIR '25. Conference Ranks: GGS A++, CORE A*. Padua, Italy: Association for Computing Machinery, July 2025, pp. 446–456. ISBN: 9798400715921. DOI: 10.1145/3726302.3730091.

19. Riccardo Lunardi, Kevin Roitero, **Michael Soprano**, Paolo Coppola, Vincenzo Della Mea, and Stefano Mizzaro. «Agent-Based Healthcare Chatbots for Regional System Services: A Case Study in the Friuli-Venezia Giulia Region». In: *Intelligent Health Systems – From Technology to Data and Knowledge*. Vol. 327. Studies in Health Technology and Informatics. Glasgow, Scotland: IOS Press, May 2025, pp. 677–681. ISBN: 978-1-64368-596-0. DOI: 10.3233/SHTI250434.

20. Francesca Da Ros, Luca Di Gaspero, Marie-Louise Lackner, Nysret Musliu, and **Michael Soprano**. «Search Trajectory Networks Applied to a Real-World Parallel Batch Scheduling Problem». In: *Applications of Evolutionary Computation (EvoApplications 2025)*. Vol. 15612. Lecture Notes in Computer Science. Conference Ranks: GGS B, CORE B. Cham: Springer Nature Switzerland, Apr. 2025, pp. 68–85. ISBN: 978-3-031-90062-4. DOI: 10.1007/978-3-031-90062-4_5.

21. **Michael Soprano**, Eddy Maddalena, Francesca Da Ros, Maria Elena Zuliani, and Stefano Mizzaro. «Evaluation of Crowdsourced Peer Review using Synthetic Data and Simulations». In: *Proceedings of the 21st Conference on Information and Research Science Connecting to Digital and Library Science*. Vol. 3937. CEUR Workshop Proceedings. Udine, Italy: CEUR-WS.org, Mar. 2025.

22. Kevin Roitero, **Michael Soprano**, David La Barbera, Eddy Maddalena, and Stefano Mizzaro. «Enhancing Fact-Checking: From Crowdsourced Validation to Integration with Large Language Models». In: *Proceedings of the 14th Italian Information Retrieval Workshop*. Vol. 3802. CEUR Workshop Proceedings. Udine, Italy: CEUR-WS.org, Oct. 2024, pp. 74–77.

23. Jaspreet Singh, **Michael Soprano**, Kevin Roitero, and Davide Ceolin. «Crowdsourcing Statement Classification to Enhance Information Quality Prediction». In: *Proceedings of the 6th Multidisciplinary International Symposium on Disinformation in Open Online Media (MISDOOM 2024)*. Lecture Notes in Computer Science. Münster, Germany: Springer Nature Switzerland, Aug. 2024, pp. 70–85. ISBN: 978-3-031-71210-4. DOI: 10.1007/978-3-031-71210-4_5.

24. **Michael Soprano**, Kevin Roitero, Vincenzo Della Mea, and Stefano Mizzaro. «Towards a Conversational-Based Agent for Health Services». In: *Proceedings of the Italia Intelligenza Artificiale - Thematic Workshops co-located with the 3rd CINI National Lab AIIS Conference on Artificial Intelligence*. Vol. 3486. CEUR Workshop Proceedings. Pisa, Italy: CEUR-WS.org, Sept. 2023, pp. 278–283.

25. David La Barbera, **Michael Soprano**, Kevin Roitero, Eddy Maddalena, and Stefano Mizzaro. «Fact-Checking at Scale with Crowdsourcing: Experiments and Lessons Learned». In: *Proceedings of the 13th Italian Information Retrieval Workshop*. Vol. 3448. CEUR Workshop Proceedings. Pisa, Italy: CEUR-WS.org, Aug. 2023, pp. 85–90.

26. Tim Draws, David La Barbera, **Michael Soprano**, Kevin Roitero, Davide Ceolin, Alessandro Checco, and Stefano Mizzaro. «The Effects of Crowd Worker Biases in Fact-Checking Tasks». In: *2022 ACM Conference on Fairness, Accountability, and Transparency*. FAccT '22. Seoul, Republic of Korea: Association for Computing Machinery, June 2022, pp. 2114–2124. ISBN: 9781450393522. DOI: 10.1145/3531146.3534629.

27. **Michael Soprano**, Kevin Roitero, Francesco Bombassei De Bona, and Stefano Mizzaro. «Crowd_Frame: A Simple and Complete Framework to Deploy Complex Crowdsourcing Tasks Off-the-Shelf». In: *Proceedings of the Fifteenth ACM International Conference on Web Search and Data Mining*. WSDM '22. Conference Ranks: GGS A+, CORE A*. Virtual Event, AZ, USA: Association for Computing Machinery, Feb. 2022, pp. 1605–1608. ISBN: 9781450391320. DOI: 10.1145/3488560.3502182.

28. Erik Brand, Kevin Roitero, **Michael Soprano**, and Gianluca Demartini. «E-BART: Jointly Predicting and Explaining Truthfulness». In: *Proceedings of The 2021 Truth and Trust Online Conference*. Virtual Event, Oct. 2021, pp. 18–27.

29. Davide Ceolin, Giuseppe Primiero, Jan Wielemaker, and **Michael Soprano**. «Assessing the Quality of Online Reviews Using Formal Argumentation Theory». In: *Web Engineering*. Conference Ranks: GGS B-, CORE B. Biarritz, France: Springer International Publishing, May 2021, pp. 71–87. ISBN: 978-3-030-74296-6. DOI: 10.1007/978-3-030-74296-6_6.

30. Kevin Roitero, **Michael Soprano**, Beatrice Portelli, Damiano Spina, Vincenzo Della Mea, Giuseppe Serra, Stefano Mizzaro, and Gianluca Demartini. «The COVID-19 Infodemic: Can the Crowd Judge Recent Misinformation Objectively?» In: *Proceedings of the 29th ACM International Conference on Information and Knowledge Management*. CIKM '20. Conference Ranks: GGS A+, CORE A. Virtual Event, Ireland: Association for Computing Machinery, Oct. 2020, pp. 1305–1314. ISBN: 9781450368599. DOI: 10.1145/3340531.3412048. Preprint: https://arxiv.org/abs/2008.05701.

31. Kevin Roitero, **Michael Soprano**, Shaoyang Fan, Damiano Spina, Stefano Mizzaro, and Gianluca Demartini. «Can The Crowd Identify Misinformation Objectively? The Effects of Judgment Scale and Assessor's Background». In: *Proceedings of the 43rd International ACM SIGIR Conference on Research and Development in Information Retrieval*. SIGIR '20. Conference Ranks: GGS A++, CORE A*. Xi'an, China (Virtual): Association for Computing Machinery, July 2020, pp. 439–448. ISBN: 9781450380164. DOI: 10.1145/3397271.3401112. Preprint: https://arxiv.org/abs/2005.06915.

32. **Michael Soprano**, Kevin Roitero, and Stefano Mizzaro. «Bias and Fairness in Effectiveness Evaluation by Means of Network Analysis and Mixture Models». In: *Proceedings of the 10th Italian Information Retrieval Workshop*. Vol. 2441. CEUR Workshop Proceedings. Padova, Italy: CEUR-WS.org, Sept. 2019, 6–7.

33. **Michael Soprano**, Kevin Roitero, and Stefano Mizzaro. «HITS Hits Readersourcing: Validating Peer Review Alternatives Using Network Analysis». In: *Proceedings of the 4th Joint Workshop on Bibliometric-enhanced Information Retrieval and Natural Language Processing for Digital Libraries co-located with the 42nd International ACM SIGIR Conference on Research and Development in Information Retrieval*. Vol. 2414. CEUR Workshop Proceedings. Paris, France: CEUR-WS.org, July 2019, 70–82.

34. **Michael Soprano** and Stefano Mizzaro. «Crowdsourcing Peer Review in the Digital Humanities?» In: *AIUCD 2019 - Book of Abstracts: Teaching and Research in Digital Humanities' Era*. Ed. by Stefano Allegrezza. AIUCD '19. Udine, Italy, Jan. 2019, pp. 251–254. ISBN: 9788894253535.

35. **Michael Soprano** and Stefano Mizzaro. «Crowdsourcing Peer Review: As We May Do». In: *Digital Libraries: Supporting Open Science*. Vol. 988. Communications in Computer and Information Science. Pisa, Italy: Springer, Jan. 2019, 259–273. DOI: {10.1007/978-3-030-11226-4_21}.

36. Kevin Roitero, **Michael Soprano**, and Stefano Mizzaro. «Effectiveness Evaluation with a Subset of Topics: A Practical Approach». In: *The 41st International ACM SIGIR Conference on Research and Development in Information Retrieval*. SIGIR '18. Conference Ranks: GGS A++, CORE A*. Ann Arbor, MI, USA: Association for Computing Machinery, June 2018, pp. 1145–1148. ISBN: 9781450356572. DOI: 10.1145/3209978.3210108.

Preprints and PhD Thesis

37. **Michael Soprano**, Andrea Cioci, and Stefano Mizzaro. *Beyond Seeing Is Believing: On Crowdsourced Detection of Audiovisual Deepfakes*. arXiv. Accepted at ROMCIR 2026, held in conjunction with ECIR 2026. May 2026. DOI: 10.48550/arXiv.2605.04797. eprint: 2605.04797 (cs.IR).

38. **Michael Soprano**. *In Crowd Veritas: Leveraging Human Intelligence to Fight Misinformation*. arXiv. Revised version of the PhD thesis, defended 22 May 2023. June 2025. DOI: 10.48550/arXiv.2506.09221. eprint: 2506.09221 (cs.IR).

Bibliometrics

Source	Publications	Citations	h-index	i10-index
Google Scholar https://scholar.google.com/citations?user=ocK0qRUAAAAJ	38	765	14	16
Scopus https://www.scopus.com/authid/detail.uri?authorId=57203392311	31	382	10	–
Web of Science (Core Collection) https://www.webofscience.com/wos/author/record/JDD-6660-2023	23	268	9	–
ACM Digital Library https://dl.acm.org/profile/99659280921	26	212	–	–

Last update: November 9, 2026

Teaching

2025/2026–2026/2027 **Information Analysis and Processing for Training** (Master's degree). Role: *Course Instructor*. Master's degree programme in Sciences and Techniques of Preventive and Adapted Physical Activities, University of Udine. 24 teaching hours in 2025/2026; 24 hours (3 CFU) in 2026/2027. In 2026/2027, Chair of the integrated course *Discipline motorie 1*, which includes this module.

2024/2025–2026/2027 **Information Analysis and Processing for Training** (Bachelor's degree). Role: *Course Instructor*. Bachelor's degree programme in Sports Science, University of Udine. 24 teaching hours per year in 2024/2025 and 2025/2026; 16 hours (2 CFU) in 2026/2027.

2024/2025–2026/2027 **Web Applications**. Role: *Course Instructor*. Master's degree programme in Management Engineering, University of Udine. Course design and teaching; 24 lectures (48 hours). Assessment based on a full-stack project and discussion.

2020/2021–2026/2027 **Social Computing**. Role: *Teaching Assistant*. Bachelor's degree programme in Internet of Things, Big Data, and Machine Learning, University of Udine. Instructor: Stefano Mizzaro. Preparation and delivery of lectures, slides, lab activities, and assignments; 1–6 lectures (2–12 hours) per academic year.

Academic Activity

Organization

2027 Reducing Online Misinformation through Credible Information Retrieval (ROMCIR 2027), ECIR 2027, Southampton, United Kingdom. **Organizer.**

2025 Workshop "AI, Misinformation and Disinformation", held as part of the 5th National CINI Conference on Artificial Intelligence (Ital-IA 2025), Trieste, June 23. **Organizer.**

21st Italian Research Conference on Digital Libraries (IRCDL 2025), Udine, February 20–21. **Website Chair.**
https://ircdl2025.uniud.it/

2024 14th Italian Information Retrieval Workshop (IIR 2024), Udine, September 5–6. **Organizer and Website Chair.**
https://iir2024.uniud.it/

2023 CARE: Conversational Agents and Responsible Ethics in Healthcare Workshop, held as part of the European Federation of Medical Informatics Special Topic Conference 2023 (EFMI STC 2023), Turin, October 27. **Organizer.**

2022 Fourth Symposium on Biases in Human Computation and Crowdsourcing (BHCC 2022), Online, October 12–13. **Proceedings and Website Chair.**

Program Committee Memberships

2026 49th International ACM SIGIR Conference on Research and Development in Information Retrieval (SIGIR 2026). **Program Committee member.**

16th Italian Information Retrieval Workshop (IIR 2026), Rome, November 12–14. **Full and Short Papers tracks.**

2025 15th Italian Information Retrieval Workshop (IIR 2025), Cagliari, September 3–5. **Full and Short Papers tracks.**

2024 12th AAAI Conference on Human Computation and Crowdsourcing (HCOMP 2024), Pittsburgh, Pennsylvania, U.S.A. **Full Papers track.**

47th International ACM SIGIR Conference on Research and Development in Information Retrieval (SIGIR 2024), Washington, D.C., U.S.A. **Full and Short Papers tracks.**

2023	45th European Conference on Information Retrieval (ECIR 2023), Dublin, Ireland. **Short Papers track.**
2022	45th International ACM SIGIR Conference on Research and Development in Information Retrieval (SIGIR 2022), Madrid, Spain. **Short Papers track.**
2021	Third Symposium on Biases in Human Computation and Crowdsourcing (BHCC 2021), Online. **Program Committee member.**

Reviewing Activities

2027	The ACM Web Conference (**WWW 2027**), Dublin, Ireland
2026	2026 Conference on Empirical Methods in Natural Language Processing (**EMNLP 2026**), Budapest, Hungary
	19th ACM International Conference on Web Search and Data Mining (**WSDM 2026**), Boise, Idaho, U.S.A.
2025	Digital Humanities (Italian Journal of Digital Humanities), ISSN 2532-8816. https://umanisticadigitale.unibo.it/
	International Journal on Digital Libraries (**IJDL**). https://portal.issn.org/resource/ISSN/0378-2484
	Data Science and Engineering (**DSEJ**). https://portal.issn.org/resource/ISSN/23641541
2023	Information Processing & Management (**IP&M**). https://portal.issn.org/resource/ISSN/0306-4573. Contribution registered on ORCID profile.
	Springer Nature Scientific Reports. https://portal.issn.org/resource/ISSN/2045-2322
	13th Italian Information Retrieval Workshop (**IIR 2023**), Pisa, Italy
2022	Information Processing & Management. https://portal.issn.org/resource/ISSN/0306-4573
	Tenth AAAI Conference on Human Computation and Crowdsourcing (**HCOMP 2022**), Online
	Conference and Labs of the Evaluation Forum (**CLEF 2022**), Bologna, Italy
	44th European Conference on Information Retrieval (**ECIR 2022**), Stavanger, Norway
2021	Third Symposium on Biases in Human Computation and Crowdsourcing (**BHCC 2021**), Delft, Netherlands (online)
	30th ACM International Conference on Information and Knowledge Management (**CIKM 2021**), Online
	32nd ACM Conference on Hypertext and Social Media (**HT 2021**), Online
2020	29th ACM International Conference on Information and Knowledge Management (**CIKM 2020**), Galway, Ireland (online)
	31st ACM Conference on Hypertext and Social Media (**HT 2020**), Online
2019	15th Italian Research Conference on Digital Libraries (**IRCDL 2019**), Pisa, Italy

Supervision and Mentoring

2020 – Pres.

Thesis supervision
3 Bachelor's (BSc) theses: 2 in Engineering and 1 in Sport Science.

Thesis co-supervision
14 Master's (MSc) and 19 Bachelor's (BSc) theses in Computer Science.

Advanced laboratory co-supervision
19 projects in Computer Science.

Counts include completed and ongoing activities.

Affiliations

2021 – Pres. **Member** of the Social, Mobile, Data & Crowd Laboratory (SMDC), University of Udine. https://smdc.uniud.it/smdc/en/members/

Honors & Awards

2025 **PhD Award 2024**. One of the two awards conferred by the University of Udine in the technical-scientific area for the best PhD theses completed in 2023. Thesis title: *In Crowd Veritas: Leveraging Human Intelligence to Fight Misinformation*.

2022 **Student Grant**. *15th ACM International Conference on Web Search and Data Mining (WSDM 2022)*, Phoenix, U.S.A. Conference registration waiver provided by WSDM. Held remotely due to the COVID-19 pandemic.

2020 **Student Grant**. *29th ACM International Conference on Information and Knowledge Management (CIKM 2020)*, Galway, Ireland. Conference registration waiver provided by CIKM. Held remotely due to the COVID-19 pandemic.

2020 **Student Volunteer**. *43rd International ACM SIGIR Conference on Research and Development in Information Retrieval (SIGIR 2020)*, Xi'an, China. Conference registration waiver provided by SIGIR. Held remotely due to the COVID-19 pandemic.

Links

ORCID	https://orcid.org/0000-0002-7337-7592
GitHub	https://github.com/Miccighel
LinkedIn	https://www.linkedin.com/in/michaelsoprano/
X	https://x.com/Miccighel_
Google Scholar	https://scholar.google.com/citations?user=ocK0qRUAAAAJ
Scopus	https://www.scopus.com/authid/detail.uri?authorId=57203392311
ACM Digital Library	https://dl.acm.org/profile/99659280921
Web of Science	https://www.webofscience.com/wos/author/record/JDD-6660-2023
DBLP	https://dblp.org/pid/222/1241.html
ResearchGate	https://www.researchgate.net/profile/Michael-Soprano
arXiv	https://arxiv.org/a/soprano_m_1
Kudos	https://www.growkudos.com/profile/michael_soprano
AIR/IRIS UniUD	https://air.uniud.it/cris/rp/rp72660
Semantic Scholar	https://www.semanticscholar.org/author/Michael-Soprano/51006308
OpenAlex	https://openalex.org/authors/A5028918939
OSF	https://osf.io/fmx3s/
Zenodo	https://zenodo.org/search?q=michael+soprano

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